

Nathaniel Billig

815-245-2448 | nbillig@uwyo.edu | <https://nbillig.github.io/> | [linkedin.com/in/nathaniel-billig/](https://www.linkedin.com/in/nathaniel-billig/)

EDUCATION

University of Wyoming , College of Engineering and Physical Sciences	Expected May 2027
Master of Science: Electrical Engineering	Cumulative GPA 3.9/4.0
Relevant Coursework: <i>Advanced Robotics Modeling, Robot Learning, Robot Perception, Cooperative Robotics, AI for Multiagent Systems</i>	
University of Wyoming , College of Engineering and Physical Sciences	May 2025
Bachelor of Science in Computer Engineering; Minors in Computer Science and Mathematics	Cumulative GPA 3.8/4.0
Relevant Coursework: <i>Advanced Microprocessors, HDL Digital Design, Electronics, GPU Programming for HPC, Computer Security, Neural Networks, Deep Reinforcement Learning</i>	

EXPERIENCE

Robotics Researcher — USSF SSTI Sponsored Research	Sept 2024 - Present
University of Wyoming	
- Designed and built a three-finger, tendon-driven robotic gripper for USSF-sponsored research, including CAD modeling, sensor selection and integration, and embedded software. - Designed a four-layer PCBA in Altium CircuitMaker for an ESP32-based controller, incorporating motor drivers, Hall-effect sensor interfaces, encoders, and load-cell amplifiers, taking the board from schematic design through layout and submission for manufacturing. - Tested gripper prototypes and revised finger geometry, sensor placement, and mechanical design based on grasping and sensing limitations and sponsor feedback. - Developed simulation and calibration methods to characterize gripper kinematics, joint sensing, and contact forces	
Microcontroller Project Engineer, 4th State Communications - Cheyenne, WY	Oct 2023 - Feb 2024
- Developed C/C++ Arduino firmware for UART communication and control of telecommunications transmitters - Built Python software using Linux NFS to transmit control commands between a central controller and networked relay systems	
Undergraduate Research Assistant, University of Wyoming - Laramie, WY	May 2023 - Sept 2025
- Developed computer vision applications and embedded Python/C/C++ software for Arduino and Raspberry Pi-based autonomous platforms - Instructed high school students and teachers in physical computing, programming fundamentals, and hands-on electronics	

PROJECTS

Hall Effect Robotic Joint Angle Sensing , University of Wyoming
· Embedded magnets and Hall-effect sensors in gripper joints to measure joint angles. Tested sensor placement and developed calibration curves and embedded software for real-time measurements.
Multi-Agent Continuum Arm Q-Learning , University of Wyoming
· Developed a multi-agent Q-learning framework for coordinated control of a continuum robotic arm, evaluating discrete state and action spaces for joint coordination and motion control
Robotic Gripper Modeling , University of Wyoming
· Built a MATLAB/Simscape model of the gripper, compared simulated joint motion with experimental measurements, and estimated contact forces during grasping.

HONORS & AWARDS

Crowd Favorite Poster — USSF University Consortium Symposium, 2026
Computer Engineering Student of the Year — Wyoming Engineering Society, 2025
Tau Beta Pi Engineering Honor Society — Inducted 2024

SKILLS

Programming & Embedded Systems: C/C++, Embedded Programming, Python, MATLAB, Arduino, Raspberry Pi, ESP32
Electronics & Communication: UART, SPI, I ² C, PCB/Schematic Design, Sensor Integration
Robotics & Simulation: ROS, Kinematics, Path Planning, Digital Twins, Reinforcement Learning, Computer Vision
Software & CAD: SolidWorks, Fusion 360, CircuitMaker, OpenCV, MATLAB/Simulink, Simscape, Microsoft Office Suite